



## DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

# POWER GENERATING SYSTEM USING FOOT-STEPS

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## **ABSTRACT**

Power Generating System Using Footsteps generates electricity by using the pressure of human footsteps. This system works on the principle of converting mechanical energy into electrical energy using sensors such as piezoelectric sensors or mechanical arrangements. When a person walks on the platform, mechanical energy is converted into electrical energy using sensors.

The generated power can be stored and used for small applications like LED lights. This system is eco-friendly, cost-effective, and useful in crowded places such as railway stations and malls. It helps in promoting renewable energy and energy conservation.

# Introduction

- The **Power Generating System Using Footsteps** is a renewable energy technology that produces electricity from human walking.
- It works on the principle of **piezoelectric effect**, where mechanical pressure is converted into electrical energy.
- When a person steps on a specially designed tile, the pressure applied activates piezoelectric sensors placed beneath the surface.
- These sensors generate electrical voltage, which is then stored in a battery for later use.
- It is an eco-friendly and non-conventional energy source that helps reduce dependence on fossil fuels.
- Though it generates small amounts of power, it is useful for lighting systems, LED displays, and charging small electronic devices.

# WORKING PRINCIPLE

- ➔ The system works on the principle of **conversion of mechanical energy into electrical energy**.
- When a person walks on the footstep platform:
- The pressure is applied on the platform creates mechanical force.
- This force activates **piezoelectric sensors** (or a mechanical arrangement like springs and gears).
- The sensors convert the mechanical energy into electrical energy.
- The generated electricity is passed through a rectifier circuit.
- The power is stored in a battery.
- The stored energy is used to operate small devices like LED lights.
- Thus, the energy produced from human footsteps is utilized to generate eco-friendly electricity.

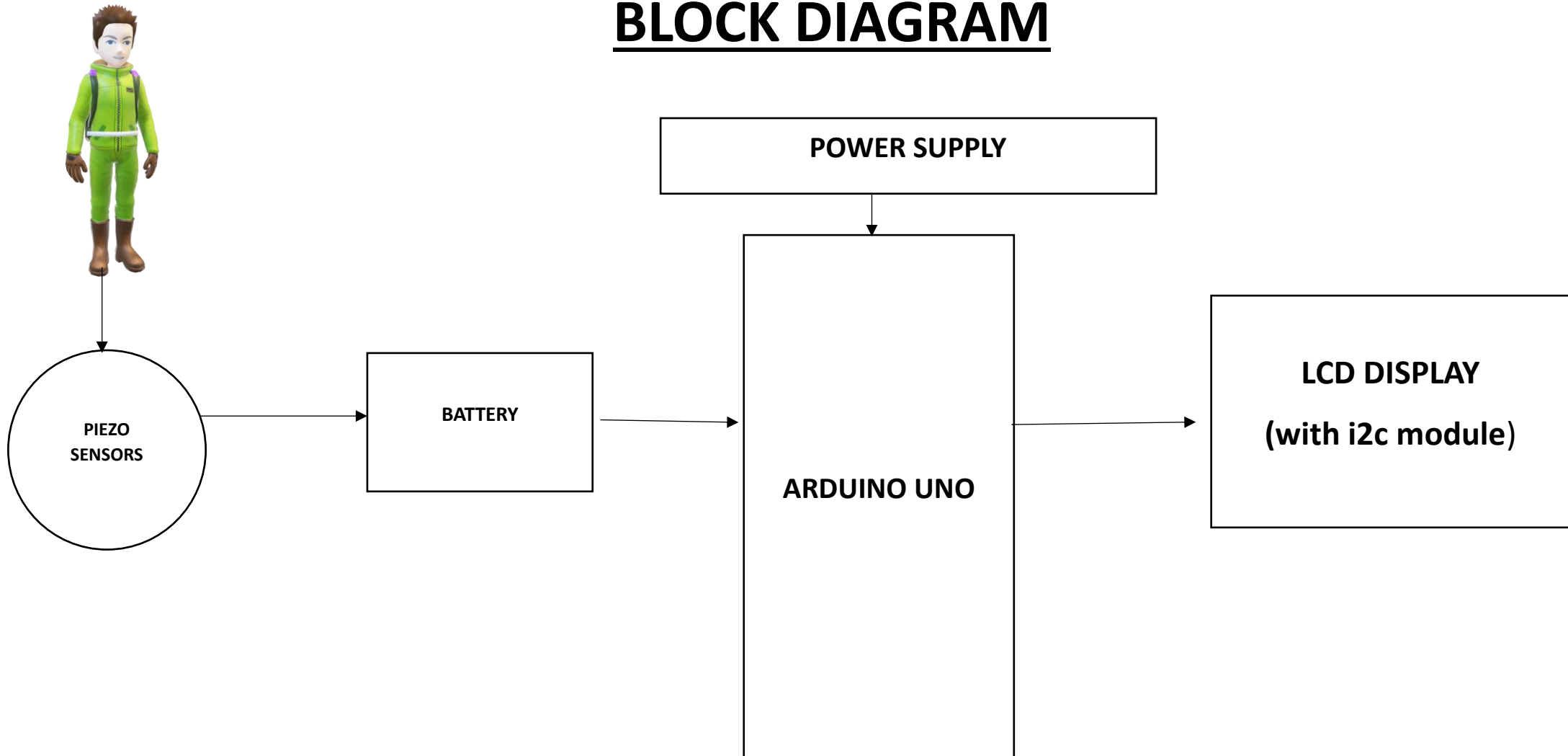
# **COMPONENTS USED**

## **HARDWARE COMONENTS:**

1. Piezo Sensors
2. Resistors
3. Diodes
4. Capacitor
5. Transistor
6. Arduino UNO
7. LCD display with I2C module
8. Jumper wires
9. Bread board

**SOFTWARE USED:** Arduino IDE

# BLOCK DIAGRAM



# Explanation of Block Diagram

## 1. Piezoelectric Sensor

- It converts the pressure of footsteps into electrical energy.
- When a person steps on it, voltage is generated.

## 2. Battery

- The generated electrical energy is stored in the battery.
- It supplies power to the system when needed.

## 3. Arduino Uno

- It is the main controller of the system.
- It processes the voltage data received from the voltage sensor.
- It controls the display section.

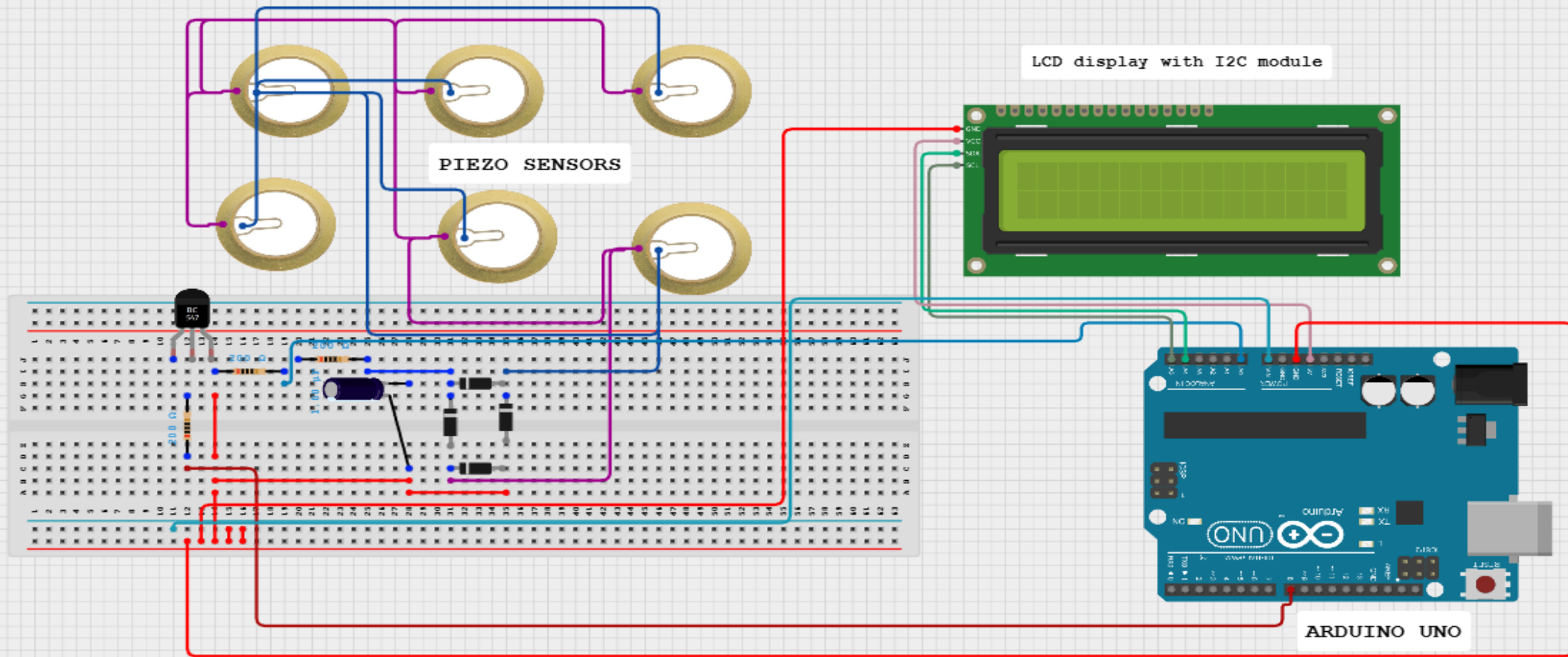
## 4. LCD Display

It displays the generated voltage or power value.

It shows how much energy is produced by footsteps.



# WIRING DIAGRAM



## APPLICATIONS

Railway stations

Bus stands

Shopping malls

Airports

Schools and colleges

Public parks

Footpaths in crowded areas

Street lighting systems

Exhibition centers

## ADVANTAGES

- Uses renewable energy (human walking energy).
- Eco-friendly and pollution free.
- Reduces electricity consumption.
- Simple and easy to install.
- Low maintenance cost.
- Useful in crowded places (railway stations, malls, schools).
- Promotes energy conservation.
- Can be used for small applications like LED lights and displays.

## **DISADVANTAGES**

- ➔ Generates small amount of power per step.
- ➔ Not suitable for large power requirements.
- ➔ Initial installation cost can be high.
- ➔ Requires heavy crowd movement to produce more energy.
- ➔ Sensors may wear out over time.
- ➔ Efficiency is lower compared to other renewable sources.
- ➔ Power output is not constant.

## CONCLUSION

The Power Generating System Using Footsteps demonstrates a practical method of converting wasted human energy into useful electrical energy. This project highlights the importance of utilizing renewable energy sources in daily life. Although the power generated from a single step is small, it becomes significant in crowded areas. The system is environmentally friendly, cost-effective in the long term, and suitable for smart city applications. With further improvements, this technology can contribute to sustainable energy solutions and reduce dependency on conventional power sources.

THANK YOU

